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§2. Cosets, quotient groups, homomorphism

Cosets & quotient groups

DEF Let H be a subgroup of G

A left coset is a set in the form $gH := \{gh \mid h \in H\}$ for some $g \in G$.

A right coset $Hg := \{hg \mid h \in H\}$ for some $g \in G$.

in particular, if $g \in H$, then $gH = Hg = H$.

REM if G is abelian, then left coset = right coset.

Write $g+H$ under the additive convention.

PROP Two left cosets g_1H, g_2H are either equal $\Leftrightarrow g_1^{-1}g_2 \in H$

or disjoint $\Leftrightarrow g_1^{-1}g_2 \notin H$

Pf. Need to show $g_1H \cap g_2H \neq \emptyset \xrightarrow{1)} g_1^{-1}g_2 \in H \xrightarrow{2)} g_1H = g_2H$
obvious

1) if $g_1H \cap g_2H \neq \emptyset$

$$\cup \\ x = g_1h_1 = g_2h_2$$

$$g_1 = xh_1^{-1} \quad g_2 = xh_2^{-1}$$

$$g_1^{-1}g_2 = h_1x^{-1}xh_2^{-1} = h_1h_2^{-1} \in H$$

2) if $g_1^{-1}g_2 \in H \Rightarrow g_1H = g_1(g_1^{-1}g_2H) = g_2H$. $\square$

DEF Write $G/H := \{gH \mid g \in G\}$ the set of left cosets

Similarly $H \backslash G := \{Hg \mid g \in G\}$.

PROP $\Rightarrow G = \bigsqcup_{gH \in G/H} gH$ disjoint union

Define $[G:H] := \#(G/H)$

the index of H as a subgroup of G .

我们想要定义出“商群”的概念,
($G/H \neq H \backslash G$)
自然希望商完的结果是唯一的,

而且构成一个群.

最自然的定义: $(aH)(bH) = abH$

要使其良定,只能是正规子群.

(证: $\forall h \in H, g \in G,$

THM (Lagrange's theorem)

if G is a finite group, $H < G$, then $\#H \mid \#G$

Pf. $[G:H] = \frac{\#G}{\#H}$ $\square$ 注意到 $\#H = \#gH$, 因为 $H \rightarrow gH, h \mapsto gh$ 是双射 $\left[\begin{array}{l} g^{-1}hg \in H \text{ 故 } \forall g \in G, g^{-1}Hg \subseteq H \\ g \text{ 换成 } g^{-1}, \text{ 可得 } H \subseteq g^{-1}Hg \\ \text{则 } g^{-1}Hg = H, \forall g \in G. \end{array} \right.$

COR 1) if G is a finite group, $x \in G$, then $\# \langle x \rangle \mid \#G$

$$2) x^{\#G} = e$$

eg $G = (\mathbb{Z}/n\mathbb{Z})^\times := (\{a \bmod n \mid \gcd(a,n)=1\}, \cdot)$ $\#G = \varphi(n)$

Then $\forall a \in (\mathbb{Z}/n\mathbb{Z})^\times, a^{\varphi(n)} \equiv 1 \pmod{n}$ (Euler's theorem)

COR If $\#G = p$ is prime, then G is cyclic ($\Rightarrow$ abelian)

Pf take $e_G \neq x \in G$ then $\# \langle x \rangle \mid \#G = p \Rightarrow \# \langle x \rangle = p \Rightarrow G = \langle x \rangle$ $\square$

DEF Let G be a group, $a, g \in G$, $\boxed{gag^{-1}}$ is called the conjugate of a by g .

LEM/DEF If $H < G, g \in G$, then $gHg^{-1} := \{ghg^{-1} \mid h \in H\}$

is a subgroup of G , called the conjugate of H by g .

Pf Let $gag^{-1}, gbg^{-1} \in gHg^{-1}, ab \in H$

$$(gag^{-1})(gbg^{-1})^{-1} = gab^{-1}g^{-1} \in gHg^{-1} \quad \square$$

DEF 1) A subgroup $H < G$ is normal if $\forall g \in G, gHg^{-1} = H$

ie all conjugates of H are H itself.

Equivalently, $\forall g \in G, gH = Hg$ (left coset = right coset)

as sets (every gh can be written as $h'g$ and vice versa)

We write $H \triangleleft G$ to indicate normal subgroups.

eg $\{e\} \triangleleft G, G \triangleleft G$

$\langle r \rangle \triangleleft D_{2n}$ but $\langle s \rangle \triangleleft D_{2n}$ not normal for $n \geq 3$

2) If $\boxed{H \triangleleft G}$, then $\forall a, b \in G, aH \cdot bH = a \cdot b \cdot H \cdot H = abH$

$a(Hb)H \rightarrow$ 并非只是形式上的操作, 是可以写出具体元素验证的.

ie we obtain a well-defined group structure on G/H .

(identity is $eH=H$, inverse of aH is $a^{-1}H$)

We call G/H the quotient group of G by H .

Group homomorphism

DEF Let $(G, \cdot)$, $(H, \circ)$ be groups.

A map $\phi: G \rightarrow H$ is called a homomorphism if

$$\forall a, b \in G, \phi(a \cdot b) = \phi(a) \circ \phi(b).$$

$$\left(\begin{array}{ll} \Rightarrow \phi(e_G \cdot e_G) = \phi(e_G) \circ \phi(e_G) & e_H = \phi(e_G) = \phi(a \cdot a^{-1}) = \phi(a) \circ \phi(a^{-1}) \\ \phi(e_G) & \Rightarrow \phi(e_G) = e_H \quad \Rightarrow \phi(a^{-1}) = (\phi(a))^{-1} \end{array} \right)$$

in particular, a bijective homomorphism is an isomorphism.

eg. $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$

$$a \mapsto a \bmod n$$

if $H \triangleleft G$ normal subgroup, then $\exists$ natural surjective homomorphism

$$\pi: G \rightarrow G/H$$

$$g \mapsto gH$$

DEF For a group homomorphism $\phi: G \rightarrow H$,

the kernel is $\ker \phi := \{g \in G \mid \phi(g) = e_H\}$.

LEM 1) The image $\phi(G)$ is a subgroup of H .

2) The kernel $\ker \phi$ is a normal subgroup of G .

Pf 1) $\forall a, b \in G, \phi(a)\phi(b)^{-1} = \phi(a)\phi(b^{-1}) = \phi(ab^{-1}) \in \phi(G)$

$$2) \forall a, b \in \ker \phi, \phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = e_H e_H^{-1} = e_H$$

$$\Rightarrow ab^{-1} \in \ker \phi \Rightarrow \ker \phi \text{ is a subgroup.}$$

$$\forall g \in G, a \in \ker \phi, \phi(gag^{-1}) = \phi(g)\underbrace{\phi(a)}_{=e_H}\phi(g)^{-1} = e_H$$

$$\Rightarrow gag^{-1} \in \ker \phi \Rightarrow \ker \phi \triangleleft G \quad \square$$

LEM A group homomorphism $\phi: G \rightarrow H$ is injective $\Leftrightarrow \ker \phi = \{e_G\}$.

Pf. " $\Rightarrow$ ": injective $\Rightarrow \ker \phi$ has 1 element. but $e_G \in \ker \phi \Rightarrow \ker \phi = \{e_G\}$.

" $\Leftarrow$ ": Suppose $\ker \phi = \{e_G\}$ and $\phi(a) = \phi(b)$ for some $a, b \in G$.

Then $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = e_H \Rightarrow ab^{-1} \in \ker \phi = \{e_G\} \Rightarrow a = b$. $\square$

REM How to construct group homomorphism?

Suppose $G = \langle S_1, \dots, S_n \mid R_1, \dots, R_m \rangle$

then to specify a homomorphism $\phi: G \rightarrow H$,

it suffices to give $\phi(S_1), \dots, \phi(S_n) \in H$, st. $\phi(R_j) = e_H, \forall j$.

eg. Determine all homomorphisms

$\phi: D_{2n} \rightarrow \mathbb{C}^\times = (\mathbb{C} \setminus \{0\}, \cdot)$

$\langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$

同态应保持 $srsr = 1$.

Find $\phi(r), \phi(s) \in \mathbb{C}^\times$, st. $\phi(r)^n = \phi(s)^2 = 1$ and $\phi(s)\phi(r)\phi(s)\phi(r) = 1$.

$$\phi(s)^2 \phi(r)^2 = 1 \Leftrightarrow \phi(r)^2 = 1$$

If n is odd, $\phi(r) = 1, \phi(s) = \pm 1$. 2 possibilities.

even $\pm 1 \quad \pm 1 \quad 4$

Some preparations for next time

PROP Let H, K be subgroups of G , define $HK := \{hk \mid h \in H, k \in K\}$. 不一定是子群

When G is finite, $\#(HK) = \frac{\#H \cdot \#K}{\#(H \cap K)}$.



Pf. HK is a disjoint union.

$HK = h_1K \sqcup h_2K \sqcup \dots \sqcup h_nK$

suffices to show for the same $h_1, \dots, h_n$,

$$H = h_1(H \cap K) \sqcup h_2(H \cap K) \sqcup \dots \sqcup h_n(H \cap K)$$

$$\left(\Rightarrow \frac{\#(HK)}{\#K} = n = \frac{\#H}{\#(H \cap K)} \right)$$

indeed, $\forall h, h' \in H, hK = h'K \Leftrightarrow h^{-1}h' \in K \Leftrightarrow h^{-1}h' \in H \cap K$

——对应

$$\Leftrightarrow h(H \cap K) = h'(H \cap K)$$

$$\text{So } HK = \bigcup_{h \in H} hK = h_1K \sqcup \dots \sqcup h_nK$$

$$\text{corresponds to } H = \bigcup_{h \in H} h(H \cap K) = h_1(H \cap K) \sqcup \dots \sqcup h_n(H \cap K) \quad \square$$

in fact, this shows a natural bijection of sets.

$$HK/K \xrightarrow{\sim} H/H \cap K$$

set of left cosets
 $hK \mapsto h(H \cap K)$

REM HK may not be a group.

eg $G = D_6$, $H = \langle s_1 \rangle$, $K = \langle s_2 \rangle$, $H \cap K = \{1\}$

$$\text{then } \#(HK) = \#H \cdot \#K = 2 \cdot 2 = 4 \neq 6$$

$\Rightarrow HK$ is not a subgroup of D_6 .

LEM If $HK \stackrel{\downarrow}{=} KH$ as sets, then HK is a subgroup of G .

eg If K is a normal subgroup,

then $\forall h \in H, hK = Kh \Rightarrow HK = KH$.

Pf. $\forall h_1, h_2 \in H, k_1, k_2 \in K$ $HK = KH$

$$h_1 k_1 (h_2 k_2)^{-1} = h_1 k_1 k_2^{-1} h_2^{-1} \stackrel{\downarrow}{=} h_1 h' k' \in HK \quad \square$$

LEM If both H, K are normal subgroups of G , then so is HK .

Pf. $\forall g \in G, gHK = HgK = HKg \quad \square$

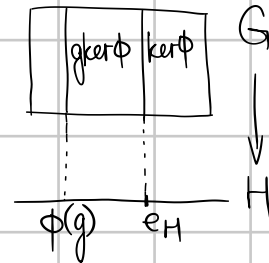
Special case: $G = HK = KH$
 $hk \rightarrow (hk)^{-1} = k^{-1}h^{-1}$

$$\Rightarrow \#G \cdot \#(H \cap K) = \#H \cdot \#K$$

§3. Isomorphism theorems, composition series, Hölder program

ITHM (First isomorphism theorem)

If $\phi: G \rightarrow H$ is a group homomorphism,
 then $\ker \phi \triangleleft G$ and $G/\ker \phi \simeq \phi(G)$.



Pf. Define a map $\psi: G/\ker \phi \rightarrow \phi(G)$

$$g\ker \phi \mapsto \phi(g)$$

1) ψ is well-defined.

$$\begin{aligned} \text{if } g_1\ker \phi = g_2\ker \phi &\Rightarrow g_1^{-1}g_2 \in \ker \phi \\ &\Rightarrow \phi(g_2) = \phi(g_1g_1^{-1}g_2) = \phi(g_1)\phi(g_1^{-1}g_2) = \phi(g_1) \end{aligned}$$

2) ψ is surjective: obvious.

3) ψ is a homomorphism.

$$\psi(g_1\ker \phi \cdot g_2\ker \phi) = \psi(g_1g_2\ker \phi) = \phi(g_1g_2) = \phi(g_1)\phi(g_2) = \psi(g_1\ker \phi)\psi(g_2\ker \phi)$$

4) ψ is injective. Suffices to check $\ker \psi \simeq \{e\}$.

$$\text{indeed, } \psi(g\ker \phi) = e_H$$

$$\parallel \phi(g) \Rightarrow g \in \ker \phi \Rightarrow g\ker \phi = \ker \phi \quad \square$$

